

1(a)	<u>force</u> proportional to <u>product</u> of masses and inversely proportional to square of separation	B1
	idea of <u>force</u> between <u>point</u> masses	B1
1(b)	mass of Jupiter (M) = $(4/3)\pi R^3 \rho$	B1
	$\omega = 2\pi/T$ or $v = 2\pi nR/T$	B1
	$(m)\omega^2 x = GM(m)/x^2$ or $(m)v^2/x = GM(m)/x^2$	M1
	substitution and correct algebra leading to $\rho T^2 = 3\pi n^3/G$	A1
1(c)(i)	$n = (4.32 \times 10^5)/(7.15 \times 10^4)$ or $n = 6.04$	C1
	$\rho \times (42.5 \times 3600)^2 = (3\pi \times 6.04^3)/(6.67 \times 10^{-11})$	C1
	$\rho = 1.33 \times 10^3 \text{ kg m}^{-3}$	A1
1(c)(ii)	Jupiter likely to be a gas/liquid (at high pressure) [allow other sensible suggestions]	B1